

Introduction to Database Systems

Applications & Purpose of Database Systems

LECTURE 1

Learning Objectives

By the end of this session, WHALES will be able to:

- 1 Define what a Database Management System (DBMS) is
- 2 Identify real-world applications of database systems
- 3 Compare the file-based approach with the DBMS approach
- 4 Explain the core purposes/goals a DBMS is designed to achieve

What is a Database Management System?

Foundational definitions

- **Database:** An organized, structured collection of related data stored electronically
- **DBMS:** Software that enables users to create, store, query, update, and manage databases
- **Database System:** The DBMS together with the data it manages
- **Goal:** Provide a convenient and efficient way to store, retrieve, and manage information

Applications of Database Systems

DBMS is used across virtually every industry

1

Banking

Accounts, transactions, and customer records

2

Airlines

Reservations, schedules, and ticketing

3

Universities

Student records, courses, grades

4

Telecom

Call records, billing, customer data

5

E-Commerce

Product catalogs, orders, inventory

6

Human Resources

Employee records, payroll, benefits

Purpose of Database Systems

Why not just use plain files?

- **Redundancy control:** Minimizes duplicate storage of the same data
- **Data integrity:** Enforces rules that keep data accurate and consistent
- **Concurrent access:** Allows many users to safely access data at the same time
- **Security:** Restricts access through authentication and authorization
- **Atomicity & recovery:** Ensures transactions complete fully or not at all, even after failures

File-Based Systems vs. DBMS

Understanding the shift in approach

File-Based System

- Data scattered across separate files
- High data redundancy & inconsistency
- Difficult concurrent access
- No built-in security or recovery
- Application-dependent data access

DBMS Approach

- Centralized, shared data repository
- Redundancy minimized via normalization
- Built-in concurrency control
- Security, integrity, and recovery mechanisms
- Data independent of any single application

Advantages of the DBMS Approach

Key benefits over traditional file processing

Advantage	Description
<u>Data independence</u>	Applications are shielded from changes in data storage structure
<u>Efficient data access</u>	Query optimizers and indexing speed up retrieval
Reduced redundancy	Centralized storage avoids duplicate data
Improved integrity	Constraints enforce accurate, consistent data
Better security	Fine-grained access control per user/role

Recap & Quick Check

Introduction to Database Systems

Key Takeaways

- A DBMS is software for creating, storing, and managing structured data
- Database systems are used across banking, airlines, retail, HR, and more
- DBMS overcomes redundancy, integrity, and concurrency problems of file systems
- Core purposes: integrity, security, concurrency, and recovery

Quick Check

Q1. Name three real-world applications of database systems.

Q2. What is one key limitation of file-based data processing?

Q3. List two purposes a DBMS is designed to serve.